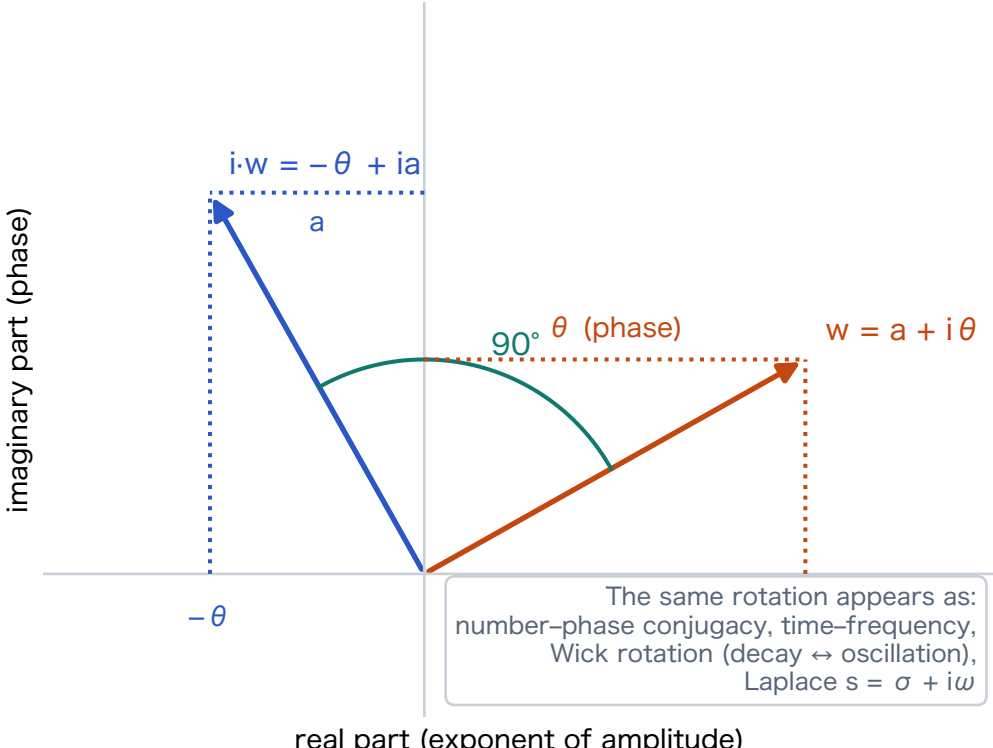
Exponent plane: multiplying by i swaps amplitude and phase (duality)
i·w = −θ + ia
a
90°
θ (phase)
w = a + iθ
imaginary part (phase)
−θ
The same rotation appears as:
number–phase conjugacy, time–frequency,
Wick rotation (decay ↔ oscillation),
Laplace s = σ + iω
real part (exponent of amplitude)